Y1 Chapter 10 Trigonometric Identities & Equations

1: Know exact trig values for $30^\circ, 45^\circ, 60^\circ$ and understand unit circle.

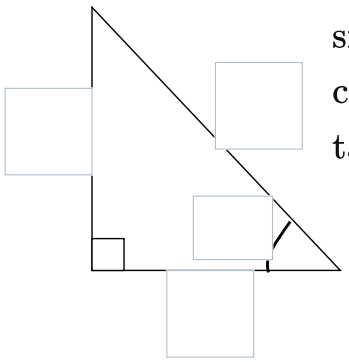
2: Use identities $\frac{\sin x}{\cos x} \equiv \tan x$ and $\sin^2 x + \cos^2 x \equiv 1$

3: Solve equations of the form $\sin(n\theta) = k$ and $\sin(\theta \pm \alpha) = k$

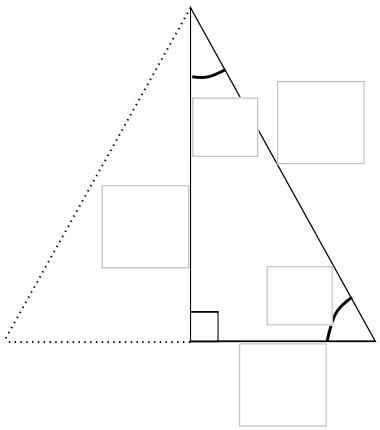
4: Solve equations which are quadratic in sin/cos/tan.

You will frequently encounter angles of $30°, 60°, 45°$ in geometric problems. Why?

Although you will always have a calculator, you need to know how to derive these.
All you need to remember:
✎ **Draw half a unit square and half an equilateral triangle of side 2.**

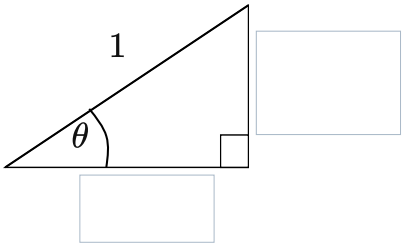


$\sin(45°) = \frac{1}{\sqrt{2}}$
$\cos(45°) = \frac{1}{\sqrt{2}}$
$\tan(45°) = 1$



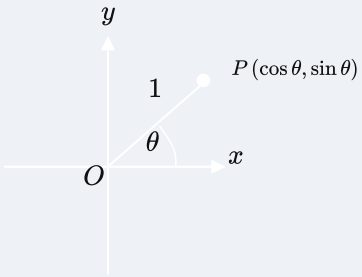
$\sin(30°) = \frac{1}{2}$
$\cos(30°) = \frac{\sqrt{3}}{2}$
$\tan(30°) = \frac{1}{\sqrt{3}}$
$\sin(60°) = \frac{\sqrt{3}}{2}$
$\cos(60°) = \frac{1}{2}$
$\tan(60°) = \sqrt{3}$

For values of θ in the range $0 < \theta < 90°$, you know that $\sin\theta$ and $\cos\theta$ are lengths on a right-angled triangle:



But how do we get the rest of the graph for sin, cos and tan when $90° \leq \theta \leq 360°$?

! The point P on a unit circle, such that OP makes an angle θ with the positive x-axis, has coordinates $(\cos\theta, \sin\theta)$. OPhas gradient $\tan\theta$.

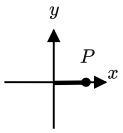
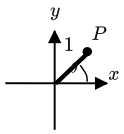


Angles are always measured **anticlockwise**.
(Further Mathematicians will encounter the same when they get to Complex Numbers)

We can consider the coordinate $(\cos\theta, \sin\theta)$ as θ increases from 0 to 360°...

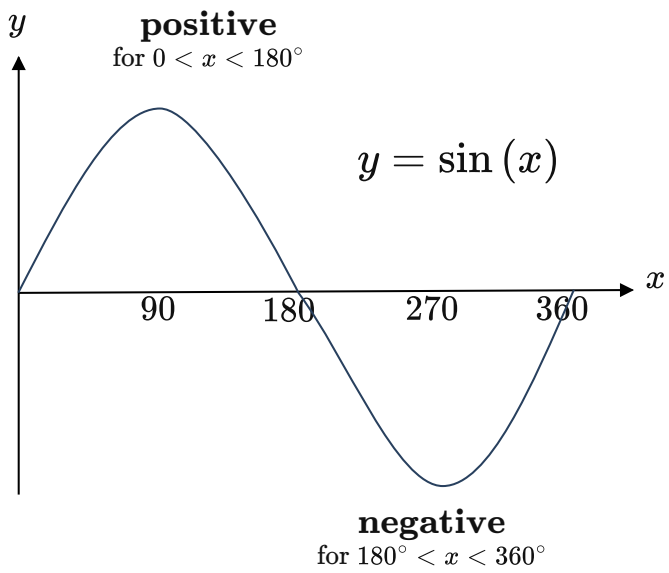
Use the unit circle to determine each value in the table, **using either "0", "+ve", "-ve", "1", "-1" or "undefined"**. Recall that the point on the unit circle has coordinate $(\cos\theta, \sin\theta)$ and OP has gradient $\tan\theta$.

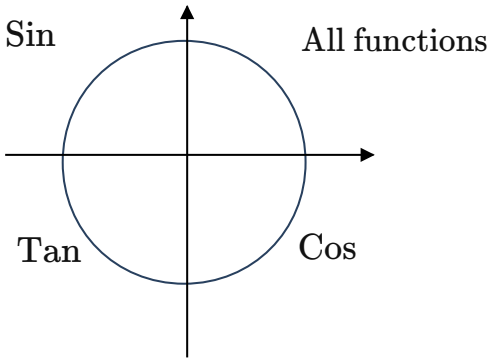
x-value y-value Gradient of OP.

	$\cos\theta$	$\sin\theta$	$\tan\theta$
$\theta = 0$ 	1	0	0
$0 < \theta < 90^\circ$ 			
$\theta = 90^\circ$			
$90^\circ < \theta < 180^\circ$			

	$\cos\theta$	$\sin\theta$	$\tan\theta$
$\theta = 180^\circ$			
$180^\circ < \theta < 270^\circ$			
$\theta = 270^\circ$			
$270^\circ < \theta < 360^\circ$			

The unit circles explains the behaviour of these trigonometric graphs beyond 90°.
However, the easiest way to remember whether $\sin(x), \cos(x), \tan(x)$ are positive or negative is to just do a **very quick sketch (preferably mentally!)** of the corresponding graph.



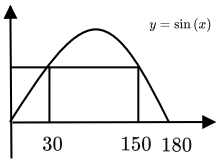


Note: The textbook uses something called '*CAST diagrams*', which is an acronym for which quadrants the **C**osine, **A**ll, **S**ine and **T**angent are positive

The following are all easily derivable using a quick sketch of a trigonometric graph, and are merely a convenience so you don't always have to draw out a graph every time. You are highly encouraged to **memorise these** so that you can do exam questions faster.

1 $\sin(x) = \sin(180^\circ - x)$

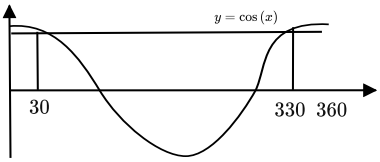
e.g. $\sin(150^\circ) = \sin(30^\circ)$



We saw this in the previous chapter when covering the 'ambiguous case' when using the sine rule.

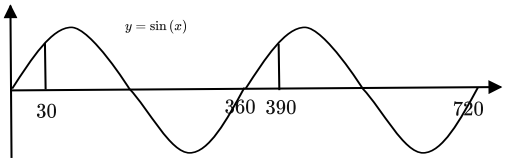
2 $\cos(x) = \cos(360^\circ - x)$

e.g. $\cos(330^\circ) = \cos(30^\circ)$



3 sin and cos repeat every 360° but tan every 180°

e.g. $\sin(390^\circ) = \sin(30^\circ)$



4 $\sin(x) = \cos(90^\circ - x)$

e.g. $\sin(50^\circ) = \cos(40^\circ)$

Remember from the previous chapter that "cosine" by definition is the sine of the "complementary" angle.
This was/is never covered in the textbook but caught everyone by surprise when it came up in a C3 exam.

Without a calculator, work out the value of each below.

$\tan(225$

$\tan(210^\circ) =$

$\sin(150^\circ) =$

$\cos(300^\circ) =$

$\sin(-45^\circ) =$

$\cos(750^\circ$

$\cos(120^\circ) =$

tan repeats every 180° so can subtract 180°

For sin we can subtract from 180°.

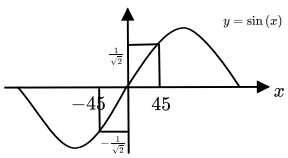
For cos we can subtract from 360°.

$\frac{1}{\sqrt{2}}$

$\frac{\sqrt{3}}{2}$

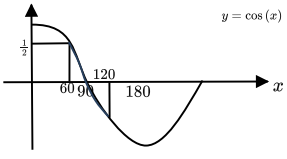
1

We have to resort to a sketch for this one.



cos repeats every 360°.

Again, let's just use a graph.



The graph is rotationally symmetric about 90°. Since 120° is 30° above 90°, we get the same y value for $90^\circ - 30^\circ = 60^\circ$, except negative.

Use the 'laws' where you can, but otherwise just draw out a quick sketch of the graph.

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- sin, cos repeat every 360° but tan every 180°

NOTE: in general, $\sin(-x) = -\sin(x)$.
Even more generally, a function f is known as an '**odd function**' if $f(-x) = -f(x)$.
tan is similarly 'odd' as $\tan(-x) = -\tan(x)$.

A function is **even** if $f(-x) = f(x)$. Examples are $f(x) = \cos(x)$ and $f(x) = x^2$ as $\cos(-x) = \cos(x)$ and $(-x)^2 = (x)^2$. Not in the exam

Without a calculator, work out the value of each below.

$\cos(315°) =$

$\sin(420°) =$

$\tan(-120°) =$

$\tan(-45°) =$

$\sin(135°) =$

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- sin, cos repeat every $360°$ but tan every $180°$

1 Express the following as trigonometric ratios of either $30°$, $45°$ or $60°$, *and hence find their exact values.* **a** $\sin 135°$ **b** $\sin(-60°)$ **c** $\sin 330°$ **d** $\sin 420°$ **e** $\sin(-300°)$ **f** $\cos 120°$ **g** $\cos 300°$ **h** $\cos 225°$ **i** $\cos(-210°)$ **j** $\cos 495°$ **k** $\tan 135°$ **l** $\tan(-225°)$ **m** $\tan 210°$ **n** $\tan 300°$ **o** $\tan(-120°)$.

Challenge

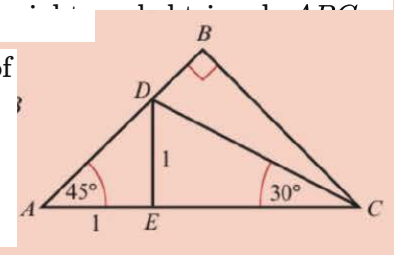
$AE = DE = 1$ unit. Angle

a Calculate the exact lengths of

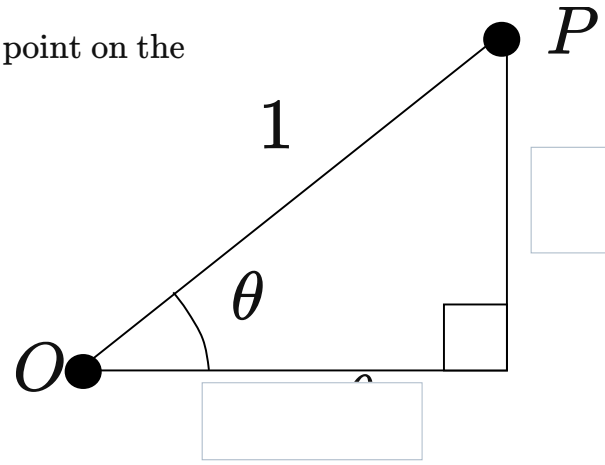
i CE **ii** DC **iii** BC **iv** DB

b State the size of angle BCD.

c Hence find exact values for



Returning to our point on the unit circle. . .



1 **Then $\tan\theta =$**

2 Pythagoras gives you...

Prove that $1 - \tan\theta \sin\theta \cos\theta \equiv \cos^2\theta$

$LHS = 1$

$= 1 -$

$= 1 -$

$= \cos^2\theta =$

$$\tan x = \frac{\sin(x)}{\cos(x)}$$
$$\sin^2 x + \cos^2 x = 1$$

Recall that $\equiv$ means 'equivalent to', and just means the LHS is **always** equal to the RHS for all values of θ. In Chapter 7 ('Proofs') we saw that usually the best method is to manipulate one side (e.g. LHS) until we get to the other (RHS).

Prove that $\tan\theta + \frac{1}{\tan\theta} \equiv \frac{1}{\sin\theta\cos\theta}$

$LHS \equiv$

$\frac{\cos^2\theta}{\sin\theta\cos\theta}$

$\equiv$

Tip #2: In any addition/subtraction involving at least one fraction (with trig functions), always combine algebraically into one.

Simplify $5 - 5\sin^2\theta$

Tip #3: Look out for $1 - \sin^2\theta$ and $1 - \cos^2\theta$. Students often don't spot that these can be simplified.

Prove that $\frac{\tan x \cos x}{\sqrt{1-\cos^2 x}} \equiv 1$

Prove that $\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} \equiv 1 - \tan^2\theta$

AQA IGCSE Further Maths Worksheet

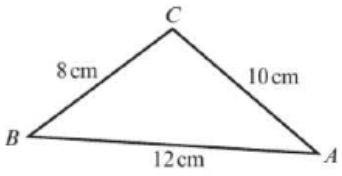
Prove that $\tan^2\theta \equiv \frac{1}{\cos^2\theta} - 1$

10 In each of the following, eliminate θ to give an equation relating x and y:

a $x = \sin\theta, y = \cos\theta$ **b** $x = \sin\theta, y = 2\cos\theta$

c $x = \sin\theta, y = \cos^2\theta$ **d** $x = \sin\theta, y = \tan\theta$

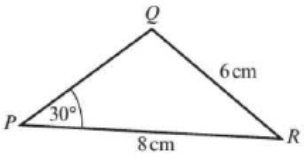
11 The diagram shows the triangle ABC with $AB = 12\,\text{cm}$, $BC = 8\,\text{cm}$ and $AC = 10\,\text{cm}$.



a Show that $\cos B = \frac{9}{16}$. (3 ma

b Hence find the exact value of $\sin B$.

12 The diagram shows triangle PQR with $PR = 8\,\text{cm}$, $QR = 6\,\text{cm}$ and angle $QPR = 30°$.



a Show that $\sin Q = \frac{2}{3}$.

b Given that Q is obtuse, find the exact value of $\cos Q$.

[MAT 2008 1C] The simultaneous equations in x, y,
$(\cos\theta)x - (\sin\theta)y = 2$
$(\sin\theta)x + (\cos\theta)y = 1$
are solvable:

A) for all values of θ in range $0 \le \theta < 2\pi$

B) except for one value of θ in range $0 \le \theta < 2\pi$

C) except for two values of θ in range $0 \le \theta < 2\pi$

D) except for three values of θ in range $0 \le \theta < 2\pi$

Remember those trigonometric angle laws (on the right) earlier this chapter? They're about to become very useful!

Reminder of 'trig laws':
- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- sin, cos repeat every 360° but tan every 180°

Solve $\sin\theta = \frac{1}{2}$ in the interval $0 \leq \theta \leq 360^\circ$.

Calculator Note:
When you do sin^{-1}, cos^{-1} and tan^{-1} on a calculator, it gives you only one value, known as the **principal value**.

Solve $5\tan\theta = 10$ in the interval $-180^\circ \leq \theta < 180^\circ$

Tip: Look out for the solution range required. $-180 \leq \theta < 180^\circ$ is a particularly common one.

tan repeats every 180°, so can add/ subtract 180° as we please.

Solve $\sin\theta = -\frac{1}{2}$ in the interval $0 \leq \theta \leq 360^\circ$.

Solve $\sin\theta = \sqrt{3}\cos\theta$ in the interval $0 \leq \theta \leq 360^\circ$.

Hint: The problem here is that we have two different trig functions. Is there anything we can divide both sides by so we only have one trig function?

Solve $2\cos\theta = \sqrt{3}$ in the interval $0 \leq \theta \leq 360°$.

Solve $\sqrt{3}\sin\theta = \cos\theta$ in the interval $-180° \leq \theta \leq 180°$.

7 A teacher asks two students to solve the equation $2\cos x = 3\sin x$ for $-180° \leq x \leq 180°$. The attempts are shown:
Student A: $\tan x = \frac{3}{2}$, $x = 56.3°$ or $x = -123.7°$.
Student B: $4\cos^2 x = 9\sin^2 x$, $4(1 - \sin^2 x) = 9\sin^2 x$, $4 = 13\sin^2 x$, $\sin x = \pm\sqrt{\frac{4}{13}}$, $x = \pm 33.7°$ or $x = \pm 146.3°$.
a Identify the mistake made by Student A. (1 mark)
b Identify the mistake made by Student B and explain the effect it has on their solution. (2 marks)
c Write down the correct answers to the question. (1 mark)

9 Find all the values of θ, to 1 decimal place, in the interval $0 < \theta < 360°$ for which $\tan^2\theta = 9$

10 a Show that $4\sin^2 x - 3\cos^2 x = 2$ can be written as $7\sin^2 x = 5$.
b Hence solve, for $0 \leq x \leq 360°$, the equation $4\sin^2 x - 3\cos^2 x = 2$.
Give your answers to 1 decimal place.

11 a Show that the equation $2\sin^2 x + 5\cos^2 x = 1$ can be written as $3\sin^2 x = 4$.
b Use your result in part a to explain why the equation $2\sin^2 x + 5\cos^2 x = 1$ has no solutions.

Harder questions replace the angle θ with a linear expression.

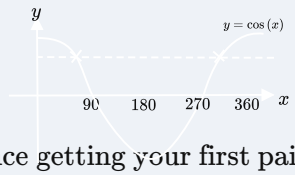
Solve $\cos 3x = -\frac{1}{2}$ in the interval $0 \leq x \leq 360°$.

STEP 1: Adjust the range of values for θ to match the expression inside the cos.

STEP 2: Immediately after applying an inverse trig function (and BEFORE dividing by 3!), find all solutions up to the end of the interval.

STEP 3: Then do final manipulation to each value.

In general you tend to get a pair of values per 360° (for any of sin/cos/tan), except for $\cos\theta = \pm 1$ or $\sin\theta = \pm 1$:



Thus once getting your first pair of values (e.g. using $\sin(180 - \theta)$ or $\cos(360 - \theta)$ to get the second value), keep adding 360° to generate new pairs.

Solve $\sin(2x + 30^\circ) = \frac{1}{\sqrt{2}}$ in the interval $0 \leq x \leq 360^\circ$.

Solve $\sin x = 2\cos x$ in the interval $0 \leq x < 300^\circ$

Solve, for $0 \leq x < 180^\circ$,
$\cos(3x - 10^\circ) = -0.4$,
giving your answers to 1 decimal place. You should show each step in your working. (7)

4 Solve for $0 \leq x \leq 180^\circ$ the equations:
a $\sin(x + 20^\circ) = \frac{1}{2}$.
b $\cos 2x = -0.8$, giving your answers to 1 decimal place.

5 a Sketch for $0 \leq x \leq 360^\circ$ the graph of $y = \sin(x + 60^\circ)$.
b Write down the exact coordinates of the points where the graph meets the coordinate axes.
c Solve, for $0 \leq x \leq 360^\circ$, the equation $\sin(x + 60^\circ) = 0.55$, giving your answers to 1 decimal place.

6 a Given that $4 \sin x = 3 \cos x$, write down the value of $\tan x$.
b Solve, for $0 \leq \theta \leq 360^\circ$, $4 \sin 2\theta = 3 \cos 2\theta$ giving your answers to 1 decimal place.

7 The equation $\tan kx = -\frac{1}{\sqrt{3}}$, where k is a constant and $k > 0$, has a solution at $x = 60^\circ$.
a Find a possible value of k.
b State, with justification, whether this is the only such possible value of k.

We saw that an equation can be 'quadratic in' something, e.g. $x - 2\sqrt{x} + 1 = 0$ is 'quadratic in $\sqrt{x}$', meaning that $\sqrt{x}$ could be replaced with another variable, say y, to produce a quadratic equation $y^2 - 2y + 1 = 0$.

Solve $5\sin^2 x + 3\sin x - 2 = 0$ in the interval $0 \leq x \leq 360°$.

Solve $\tan^2\theta = 4$ in the interval $0 \leq x \leq 360°$.

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ in the interval $-180° \leq x \leq 180°$.

Tip: We have an identity involving sin^2 and cos^2, so it makes sense to change the squared one that would match all the others.

(a) Show that the equation $5\sin x = 1 + 2\cos^2 x$ can be written in the form $2\sin^2 x + 5\sin x - 3 = 0$. (2)
(b) Solve, for $0 \leq x < 360°$, $2\sin^2 x + 5\sin x - 3 = 0$. (4)

5 Find all the solutions, in the interval $0 \leq x \leq 360°$, to the equation $8\sin^2 x + 6\cos x - 9 = 0$ giving each solution to one decimal place. (6 marks)

6 Find, for $0 \leq x \leq 360°$, all the solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ giving each solution to one decimal place.

7 Show that the equation $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

8 a Show that the equation $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

b Hence show that the equation $\cos^2 x = 2 - \sin x$ has no solutions. (3 marks)

9 $\tan^2 x - 2\tan x - 4 = 0$.

a Show that $\tan x = p \pm \sqrt{q}$ where p and q are numbers to be found.

b Hence solve the equation $\tan^2 x - 2\tan x - 4 = 0$ in the interval $0 \leq x \leq 540°$.

1

2 *[MAT 2014 1E]* As x varies over the real numbers, the largest value taken by the function $\left(4\sin^2 x + 4\cos x + 1\right)^2$ equals what?

$\left(\mathbf{4-4\cos^2 x+4\cos x+1}\right)^2$